

Notice how short this code is. The neatly formatted file *readable_eq_data.json* has over 6,000 lines. But in just a few lines, we can read through all that data and store it in a Python list. Next, we'll pull the magnitudes from each earthquake.

Extracting Magnitudes

We can loop through the list containing data about each earthquake, and extract any information we want. Let's pull out the magnitude of each earthquake:

```
eq_explore    --snip--
_data.py      all_eq_dicts = all_eq_data['features']

❶ mags = []
    for eq_dict in all_eq_dicts:
❷     mag = eq_dict['properties']['mag']
        mags.append(mag)

print(mags[:10])
```

We make an empty list to store the magnitudes, and then loop through the list `all_eq_dicts` ❶. Inside this loop, each earthquake is represented by the dictionary `eq_dict`. Each earthquake's magnitude is stored in the 'properties' section of this dictionary, under the key 'mag' ❷. We store each magnitude in the variable `mag` and then append it to the list `mags`.

We print the first 10 magnitudes, so we can see whether we're getting the correct data:

```
[1.6, 1.6, 2.2, 3.7, 2.92000008, 1.4, 4.6, 4.5, 1.9, 1.8]
```

Next, we'll pull the location data for each earthquake, and then we can make a map of the earthquakes.

Extracting Location Data

The location data for each earthquake is stored under the key "geometry". Inside the geometry dictionary is a "coordinates" key, and the first two values in this list are the longitude and latitude. Here's how we'll pull this data:

```
eq_explore    --snip--
_data.py      all_eq_dicts = all_eq_data['features']

mags, lons, lats = [], [], []
for eq_dict in all_eq_dicts:
    mag = eq_dict['properties']['mag']
❶ lon = eq_dict['geometry']['coordinates'][0]
    lat = eq_dict['geometry']['coordinates'][1]
    mags.append(mag)
    lons.append(lon)
```
